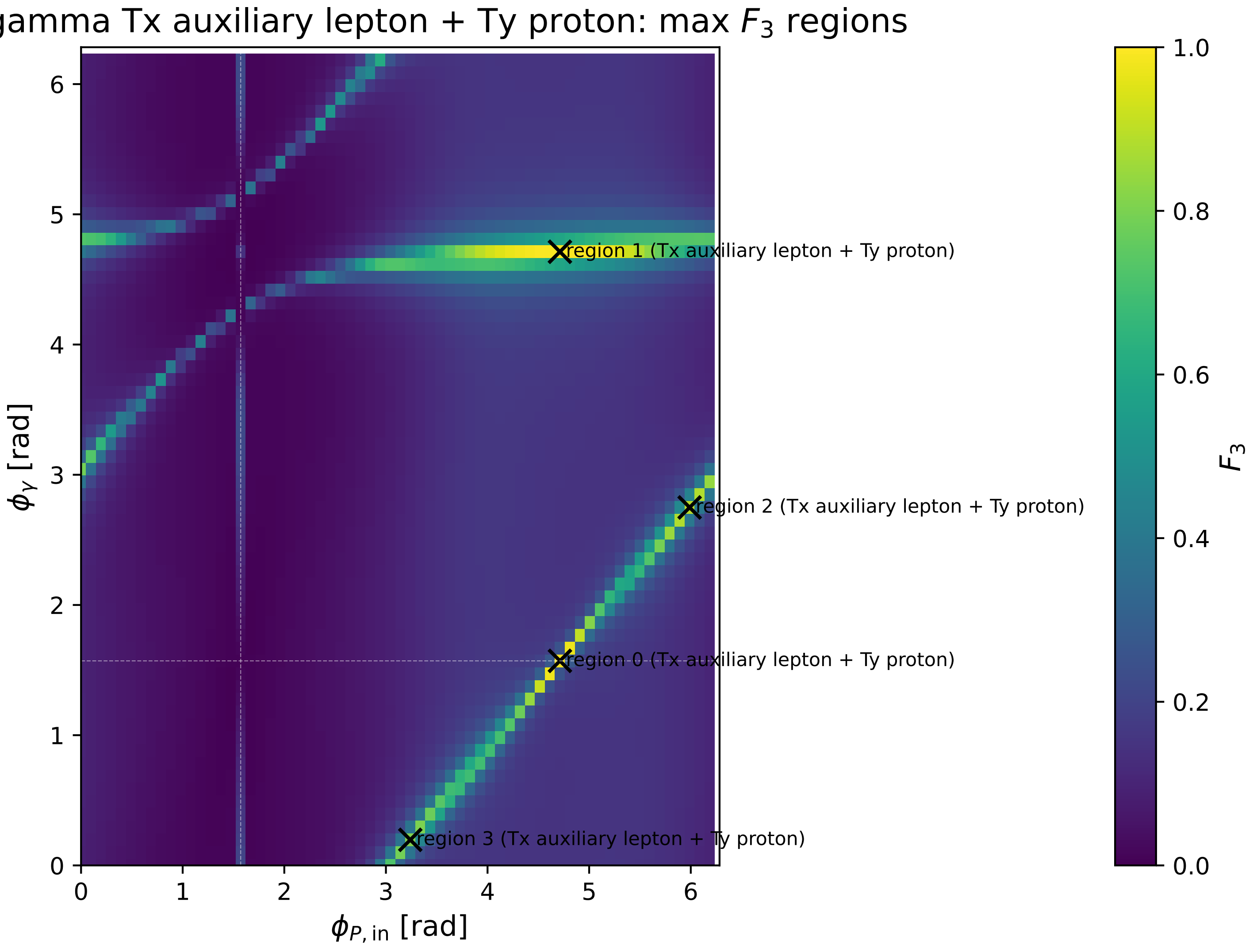
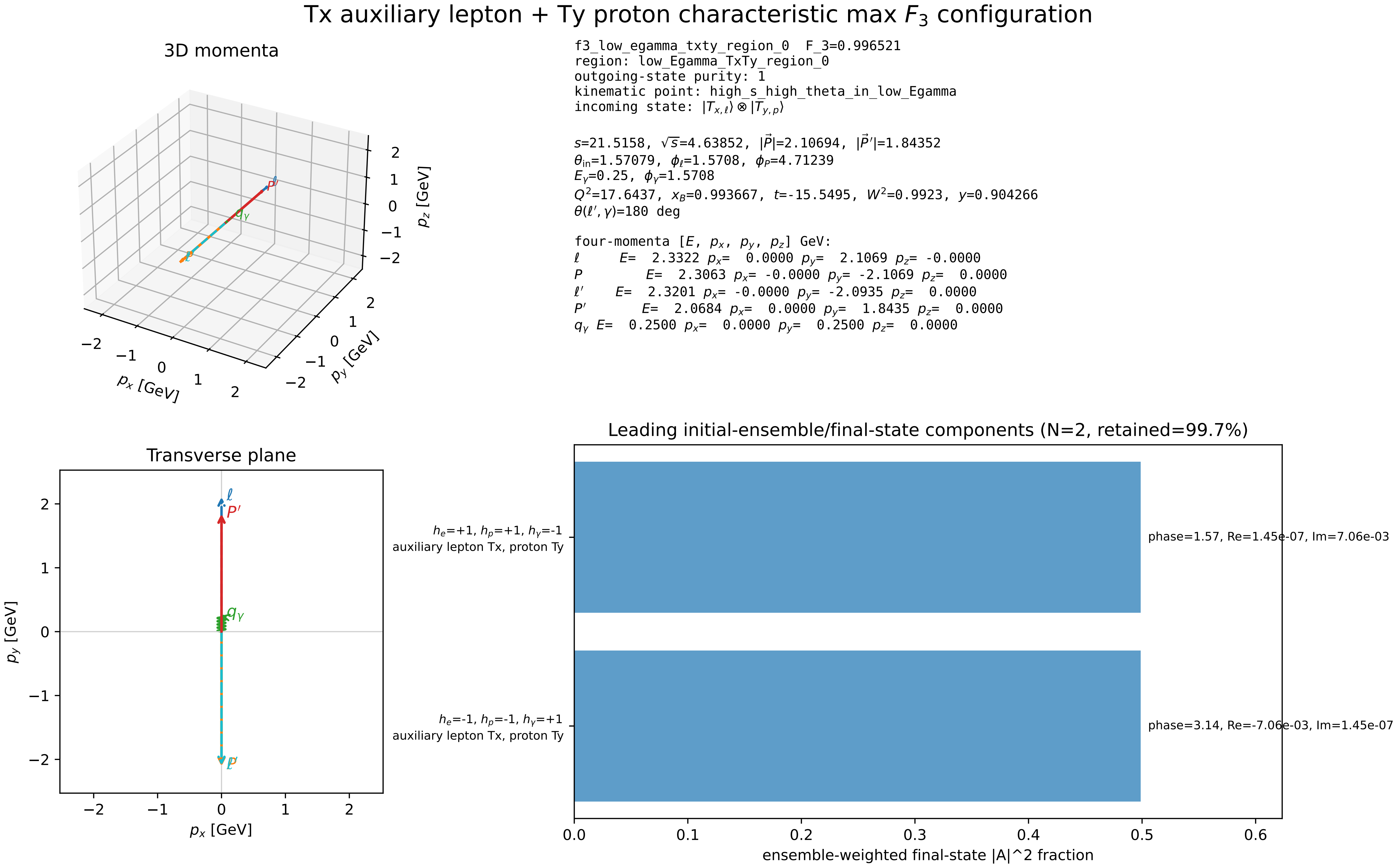
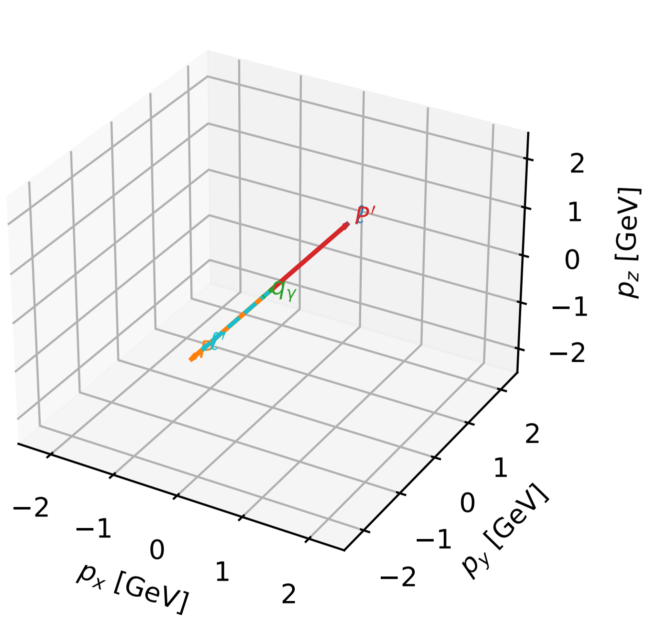


gamma Tx auxiliary lepton + Ty proton: max F_3 regions
region 1 (Tx auxiliary lepton + Ty proton)
region 2 (Tx auxiliary lepton + Ty proton)
region 0 (Tx auxiliary lepton + Ty proton)
region 3 (Tx auxiliary lepton + Ty proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
F_3
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0


Tx auxiliary lepton + Ty proton characteristic max F_3 configuration
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
ℓ
P'
q_γ
ℓ'
f3_low_egamma_txty_region_0 F_3=0.996521
region: low_Egamma_TxTy_region_0
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|T_{x,\ell}\rangle \otimes |T_{y,p}\rangle$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=1.84352
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=0.25, ϕ_γ=1.5708
Q^2=17.6437, x_B=0.993667, t=-15.5495, W^2=0.9923, y=0.904266
$\theta(\ell', \gamma)$=180 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= 0.0000 p_y= 2.1069 p_z= -0.0000
P E= 2.3063 p_x= -0.0000 p_y= -2.1069 p_z= 0.0000
ℓ' E= 2.3201 p_x= -0.0000 p_y= -2.0935 p_z= 0.0000
P' E= 2.0684 p_x= 0.0000 p_y= 1.8435 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.0000 p_y= 0.2500 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
Leading initial-ensemble/final-state components (N=2, retained=99.7%)
h_e=+1, h_p=+1, h_γ=-1
auxiliary lepton Tx, proton Ty
phase=1.57, Re=1.45e-07, Im=7.06e-03
h_e=-1, h_p=-1, h_γ=+1
auxiliary lepton Tx, proton Ty
phase=3.14, Re=-7.06e-03, Im=1.45e-07
ensemble-weighted final-state |A|^2 fraction

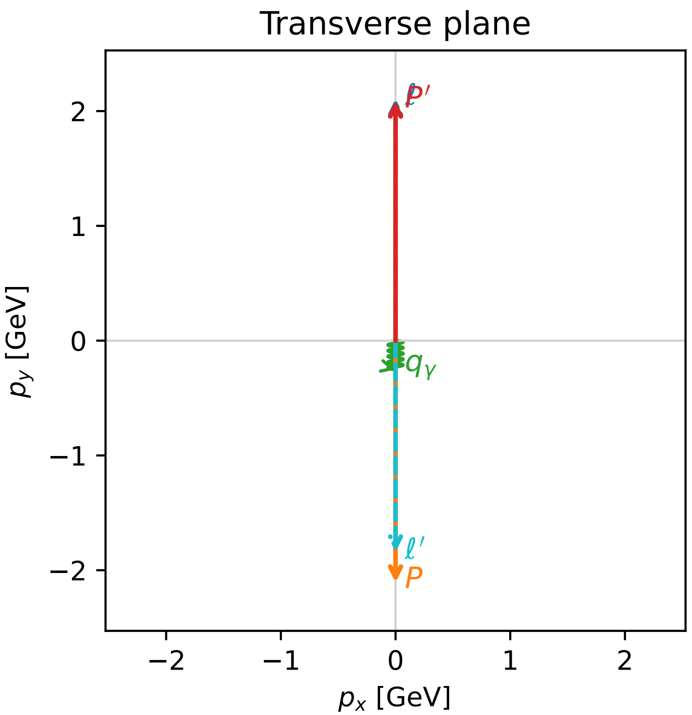
Tx auxiliary lepton + Ty proton characteristic max F_3 configuration

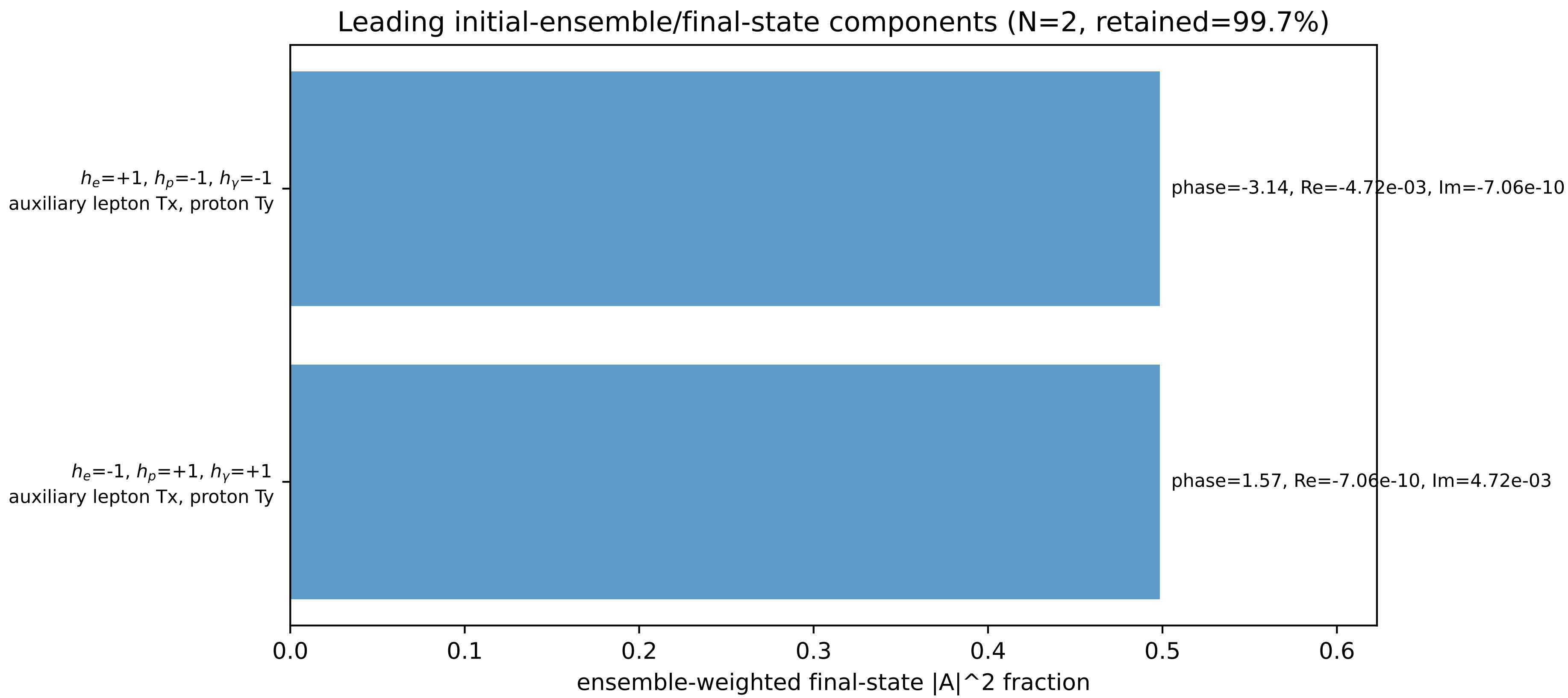


f3_low_egamma_txty_region_1 F_3=0.995627
region: low_Egamma_TxTy_region_1
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|T_{x,\ell}\rangle \otimes |T_{y,p}\rangle$

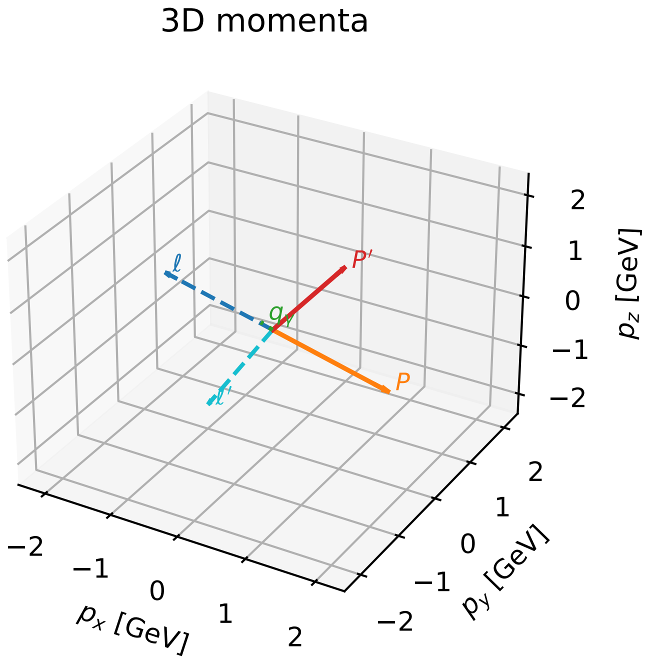
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=2.09195
θ_{in}=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=0.25, ϕ_γ=4.71239
Q^2=15.5379, x_B=0.876353, t=-17.6305, W^2=3.07213, y=0.902944
$\theta(\ell',\gamma)$=0 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= 0.0000 p_y= 2.1069 p_z= -0.0000
P E= 2.3063 p_x= -0.0000 p_y= -2.1069 p_z= 0.0000
ℓ' E= 2.0959 p_x= 0.0000 p_y= -1.8420 p_z= 0.0000
P' E= 2.2926 p_x= 0.0000 p_y= 2.0920 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.0000 p_y= -0.2500 p_z= 0.0000





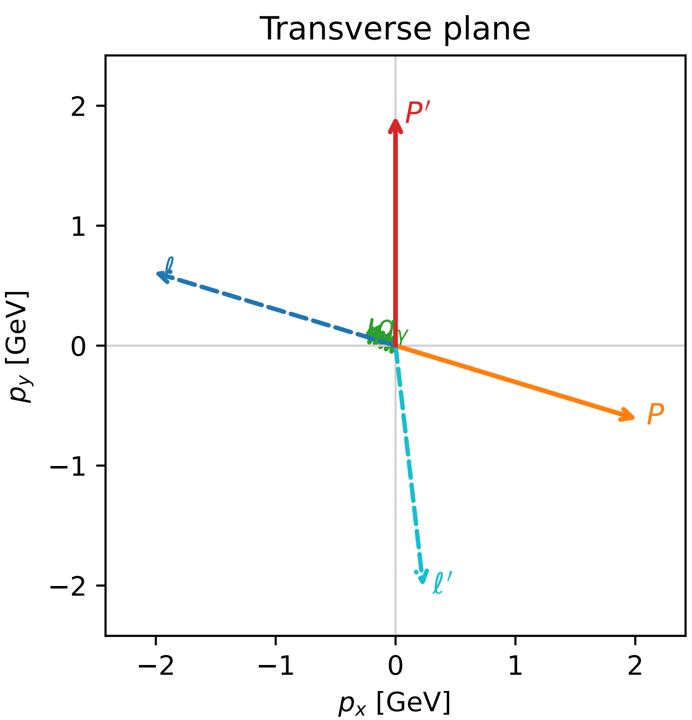
Tx auxiliary lepton + Ty proton characteristic max F_3 configuration

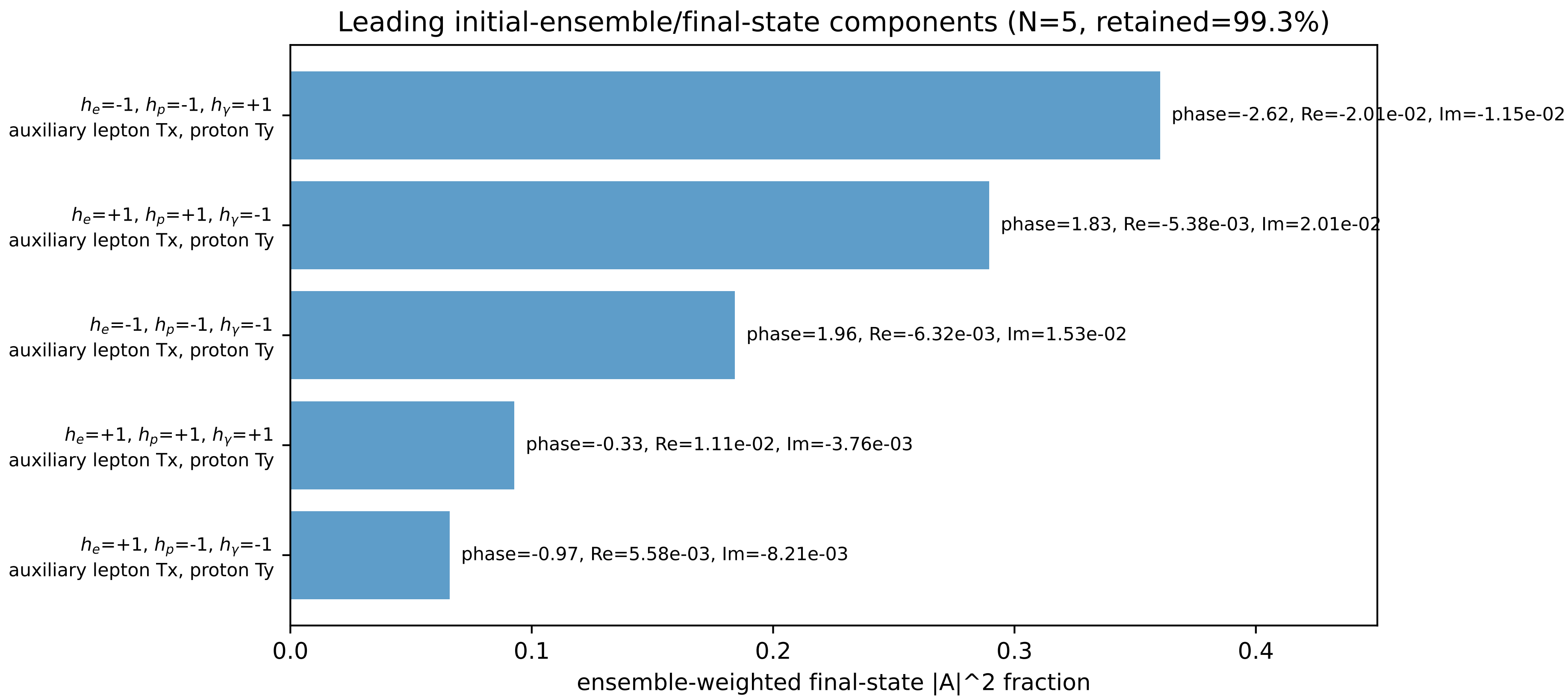


f3_low_egamma_txty_region_2 F_3=0.896875
region: low_Egamma_TxTy_region_2
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|T_{x,\ell}\rangle \otimes |T_{y,p}\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=1.91426
θ_{in}=1.57079, ϕ_ℓ=2.84707, ϕ_P=5.98866
E_γ=0.25, ϕ_γ=2.74889
Q^2=11.9166, x_B=0.944549, t=-10.4147, W^2=1.57943, y=0.642504
$\theta(\ell',\gamma)$=119.055 deg

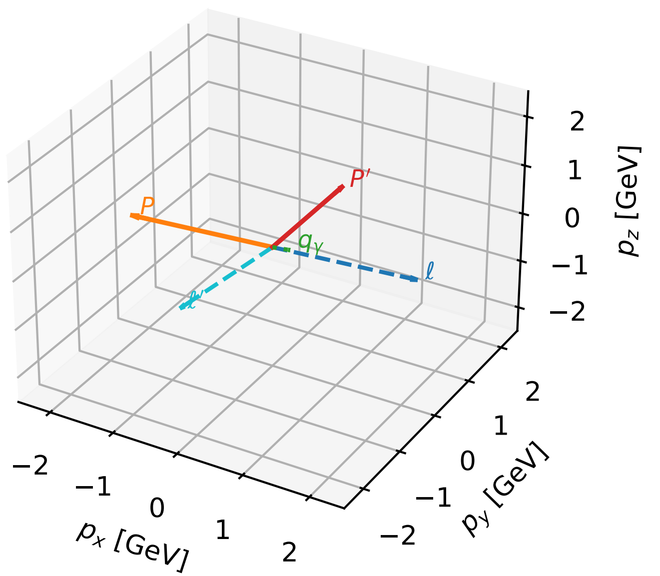
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= -2.0162 p_y= 0.6116 p_z= -0.0000
P E= 2.3063 p_x= 2.0162 p_y= -0.6116 p_z= 0.0000
ℓ' E= 2.2568 p_x= 0.2310 p_y= -2.0099 p_z= 0.0000
P' E= 2.1317 p_x= 0.0000 p_y= 1.9143 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.2310 p_y= 0.0957 p_z= 0.0000





Tx auxiliary lepton + Ty proton characteristic max F_3 configuration

3D momenta

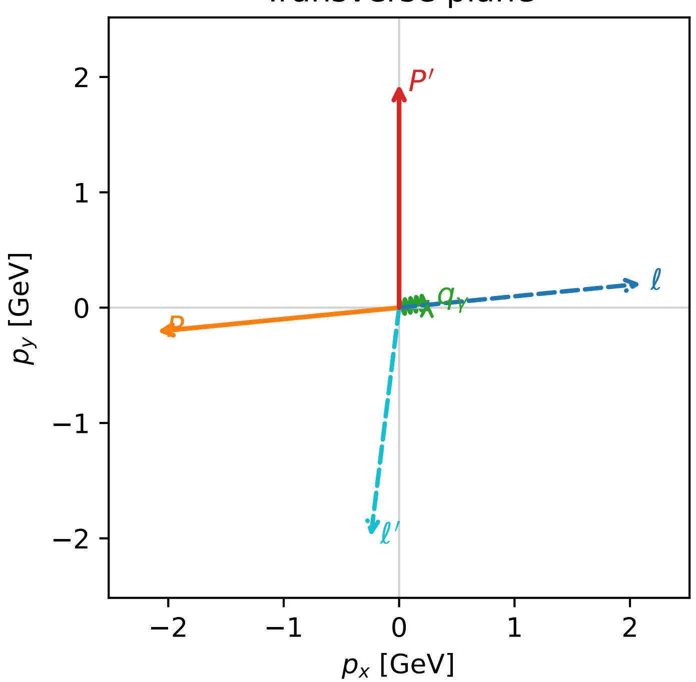


f3_low_egamma_txty_region_3 F_3=0.811565
region: low_Egamma_TxTy_region_3
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|T_{x,\ell}\rangle \otimes |T_{y,p}\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=1.93673
θ_{in}=1.57079, ϕ_ℓ=0.0981748, ϕ_P=3.23977
E_γ=0.25, ϕ_γ=0.19635
Q^2=10.2807, x_B=0.920571, t=-8.96624, W^2=1.76689, y=0.568741
$\theta(\ell', \gamma)$=108.29 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= 2.0968 p_y= 0.2065 p_z= -0.0000
P E= 2.3063 p_x= -2.0968 p_y= -0.2065 p_z= 0.0000
ℓ' E= 2.2366 p_x= -0.2452 p_y= -1.9855 p_z= 0.0000
P' E= 2.1519 p_x= 0.0000 p_y= 1.9367 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.2452 p_y= 0.0488 p_z= 0.0000

Transverse plane



Leading initial-ensemble/final-state components (N=5, retained=99.4%)